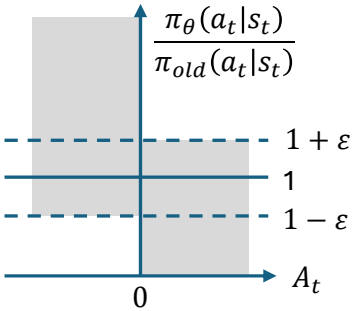
$$J^{PPO}(\theta) = E_t[\min(\frac{\pi_\theta(a_t|s_t)}{\pi_{old}(a_t|s_t)}A_t, clip(\frac{\pi_\theta(a_t|s_t)}{\pi_{old}(a_t|s_t)}, 1-\varepsilon, 1+\varepsilon)A_t)]$$

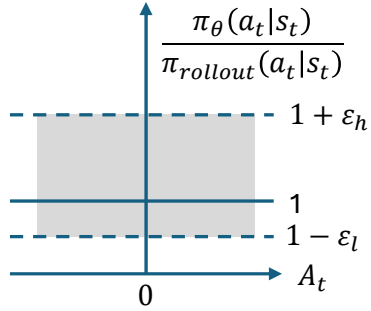


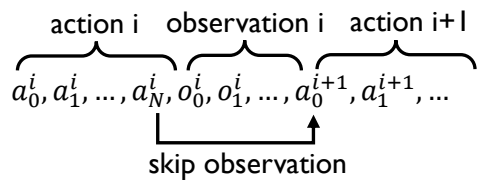

$$\hat{A}_t^{GAE} = \sum_{l=0}^{|y|-t-1} (\gamma\lambda)^l \delta_{t+l}$$

$$\Rightarrow \hat{A}_t^{GAE} = \delta_t + (\gamma\lambda)\hat{A}_{t+1}^{GAE}$$

where $\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$

$$J^{SAO}(\theta) = E_t[clip(\frac{\pi_\theta(a_t|s_t)}{\pi_{rollout}(a_t|s_t)}, 1-\varepsilon_l, 1+\varepsilon_h)A_t]$$




$$\hat{A}_{i,N}^{GAE} = \delta_{i,N} + (\gamma\lambda)\hat{A}_{i+1,0}^{GAE}$$

where $\delta_{i,N} = r_{i,N} + \gamma V_\phi(s_{i+1,0}) - V_\phi(s_{i,N})$